

Point	X	Y	Class
A	1	2	C1
B	2	3	C1
C	3	3	C2
D	6	5	C2

Classify test point **T(2,2)** using **K = 3** and Euclidean distance.

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Distances from T(2,2):

- $d(T,A) = \sqrt{[(2-1)^2 + (2-2)^2]} = \sqrt{1} = 1$
- $d(T,B) = \sqrt{[(2-2)^2 + (2-3)^2]} = \sqrt{1} = 1$
- $d(T,C) = \sqrt{[(2-3)^2 + (2-3)^2]} = \sqrt{2} \approx 1.41$
- $d(T,D) = \sqrt{[(2-6)^2 + (2-5)^2]} = \sqrt{25} = 5$
- **Nearest 3 neighbors:**
- A (C1), B (C1), C (C2)
- **Majority Class:**
- C1 → **Final Answer: T belongs to C1**

2. KNN with Different K Values

Problem 2

Using same dataset, classify T(2,2) with:

- (a) K = 1
- (b) K = 3
- (c) K = 4
- **Solution**
- Distances already computed.
- **(a) K = 1**
- Nearest: A (C1)
- **Class = C1**
- **(b) K = 3**
- Neighbors: A, B, C
- C1 = 2, C2 = 1
- **Class = C1**

- **(c) $K = 4$**
- All points considered
 $C1 = 2, C2 = 2$

Tie condition

Usually resolved by:

- Choosing smaller K
- Distance-weighted voting

3. Distance Weighted KNN

Problem 3

Same data, classify $T(2,2)$ using **$K = 3$ with weighted voting**:

Weight formula:

Point	Class	Distance	Weight
A	C1	1	1
B	C1	1	1
C	C2	1.41	0.71

Point A

Distance = 1

Point B

Distance = 1

Point C

Distance = 1.41

Step 4: Add weights CLASS-WISE

Now don't count points — **add weights**

For Class C1:

- $A \rightarrow 1$
- $B \rightarrow 1$

For Class C2:

- $C \rightarrow 0.71$

Step 5: Compare totals

Bigger weight wins

Sum of weights:

- $C1 = 1 + 1 = 2$
- $C2 = 0.71$

Final Class = C1

4. KNN in 1-D Data

Problem 4

Training data:

Value	Class
2	A
4	A
7	B
8	B

Classify test point **5** with $K = 3$.

Solution

Distances:

- $|5-2| = 3$
- $|5-4| = 1$
- $|5-7| = 2$
- $|5-8| = 3$

Nearest 3: 4(A), 7(B), 2(A)

$A = 2$, $B = 1$

Final Class = A

Manhattan Distance KNN

Problem 6

Classify T(3,3) using **Manhattan distance**, $K=3$.

Point	X	Y	Class
A	1	2	C1
B	2	1	C1
C	4	3	C2
D	5	4	C2

Solution

Manhattan formula:

Distances:

- A: $|3-1| + |3-2| = 2 + 1 = 3$
- B: $|3-2| + |3-1| = 1 + 2 = 3$
- C: $|3-4| + |3-3| = 1 + 0 = 1$
- D: $|3-5| + |3-4| = 2 + 1 = 3$

Nearest 3: C (C2), A (C1), B (C1)

C1 = 2, C2 = 1

Final Class = C1

Problem: Compare Manhattan vs Euclidean Distance

Point	X	Y	Class
A	1	1	C1
B	2	2	C1
C	4	4	C2
D	5	5	C2

Test Point:

T(3,3) K = 3

Distances:

- A $\rightarrow \sqrt{[(3-1)^2 + (3-1)^2]} = \sqrt{8} \approx \mathbf{2.83}$
- B $\rightarrow \sqrt{[(3-2)^2 + (3-2)^2]} = \sqrt{2} \approx \mathbf{1.41}$
- C $\rightarrow \sqrt{[(3-4)^2 + (3-4)^2]} = \sqrt{2} \approx \mathbf{1.41}$
- D $\rightarrow \sqrt{[(3-5)^2 + (3-5)^2]} = \sqrt{8} \approx \mathbf{2.83}$

Nearest 3 neighbors:

B (C1), C (C2), A (C1)

Count:

- C1 = 2
- C2 = 1

Result (Euclidean):

Class = C1

Step 2: Manhattan Distance

Formula:

Distances:

- A $\rightarrow |3-1| + |3-1| = 2 + 2 = \mathbf{4}$

- $B \rightarrow |3-2| + |3-2| = 1 + 1 = 2$
- $C \rightarrow |3-4| + |3-4| = 1 + 1 = 2$
- $D \rightarrow |3-5| + |3-5| = 2 + 2 = 4$

Nearest 3 neighbors:

B (C1), C (C2), A (C1)

Count:

- $C1 = 2$
- $C2 = 1$

Result (Manhattan):

Class = C1

Feature	Euclidean	Manhattan
Formula	$\sqrt{\text{squared diff}}$	Sum of absolute diff
Shape	Circular distance	Diamond/grid distance
Sensitive to outliers	High	Low
Best for	Continuous data	Grid / discrete data
Computation	Slower	Faster

Which is BEST?

No single “best” — depends on data

Use Euclidean when:

- Features are continuous
- Data is smooth
- Geometry matters (real-world distance)

Use Manhattan when:

- Data is grid-like (city blocks)
- High dimensions (faster, more stable)
- Less sensitivity to outliers needed